formance, while the E cores are built for scalable multicore performance. You put those two things together along with our Intel thread director, which is hardware baked directly into the silicon that we tie into the OS to deliver optimal performance by adapting to the workloads and optimizing the use of both the P cores and the E cores.

d You mentioned the Intel thread director, which is baked into the hardware. What about the software? How do applications, adapt to the new way of dealing with the P cores and E cores. Are you working with developers to ensure applications make optimal use of the architecture?
We've always worked with ecosystem partners on software and optimizing threading libraries. Operating systems have always made decisions on where to place threads and workloads based on information. But, mostly the information was limited to basic statistics around foreground and background tasks that the application designer tells the operating system that he wants to flag. What the Intel thread director does is, it enables these two new core architectures to work seamlessly together, by guiding the operating system. It gives it a lot more information than it ever had. From real-time state of the core to the instruction mix to various other telemetrics that go all the way down to the granular level. It allows the OS to make more intelligent scheduling decisions that it had in the past. So, that's really the big advantage with the thread director.

d Is the hybrid architecture the way forward for Intel? Will we see the upcoming generations of Intel processors adapt to the hybrid direction?
Our approach starts with performance as opposed to other companies where they start with battery savings on mobile, where they really focus on power consumption. We started with performance, our most important goal when we designed the hybrid approach was to support all client segments with a single highly scalable architecture that caters to the three design points. First, is the maximum performance in the two-chip socket on the desktop side with leadership performance while enabling power efficiency even in that design with the efficiency cores, all the way through to the high-performance mobile BGA package. When you add the larger XE graphics and Thunderbolt 4 connectivity, we can fit a lot into the mobile performance envelope. And then scaling all the way down to the thin, low-power high-density packages, where you optimise that I/O and power delivery. For us, hitting those three design points and scaling the architecture from the top to the bottom was really important. So, to answer your question, we will always focus on performance and optimise for the best experience. While we can't talk about products that haven't been announced yet, we are very comfortable with this hybrid approach, particularly the way it scales all the way down, from high performance to lower power designs.

d How does the Intel 12th gen platform compare to the competition? The AMD Ryzen 5000 series on the desktop and the Ryzen 6000 series on the mobile front?
I am not going to comment on other company's processors. What I can say though, is that we stand firmly behind our claim of the world's best gaming processor, with the Intel Core i9-12900K, that we launched back in Q4 2021. That holds firm not by just our numbers but also by the reviewers around the world. And even when you look at the mid-segment on the desktop side, going down from 16 cores at the top to the 8-12 core segment offered by the 12th gen Core i5 series, even they deliver great performance along with the great value. On the mobile side, with our announcements at the CES, we have the claim for the fastest gaming processor on the mobile platform as well. From our perspective, the 12th Gen Intel processors bring new levels of industry-leading performance and we stand behind them.

d How does the DDR5 aspect play out? Do you see more of 12th Gen-based laptops to launch with DDR5 memory this year or is it going to be a mix of DDR4 and DDR5 that we see this year? Also, what is the performance delta on the 12th gen processors when it comes to DDR4 vs DDR5 across applications?
This year, the DDR5 memory technology is in transition and the 12th generation processors do support DDR5 as well as DDR4 memory. On the desktop side the focus is on DDR5 while on the notebook side, we see more DDR4 based products. Over time, as DDR5 continues to mature, we expect the adoption to ramp up across the industry.

About the performance delta. Systems with DDR5 do show that they are faster on games, however, even with DDR4, it is still a great experience. The CPU itself is doing a whole lot of heavy lifting and the memory is a bonus. Sure, there is a performance bump with DDR5, but you still have great experiences with DDR4 on the Intel 12th gen processors.

d The Intel 12th Gen processors also bring a brand new P-series on the mobile front, where do they sit in the overall scheme of things?
The P series is in the performance space, in comparison to the U series that were focused on the smaller, thin and light series. While not being in the same league as the enthusiasts' level that the H-series caters to, the P series strikes a balance between higher-end performance and solid endurance on the battery life front. d